2022-Ph-H1-Q3
Section: Our Dynamic Universe
Topic: Motion, Equations and Graphs

A projectile has initial vertical velocity $20\,\mathrm{m\,s^{-1}}$. How long does it take to reach max height?

Answer: **B** ($2.0\,\mathrm{s}$)

Solution:
At maximum height, $v = 0$ and $a = -9.8\,\mathrm{ms^{-2}}$
Use:

$$v = u + at \Rightarrow 0 = 20 - 9.8t \Rightarrow t = \frac{20}{9.8} \approx 2.0\ \mathrm{s}$$

Guidance for Students:
Remember the vertical velocity becomes 0 at the top. The time to max height only depends on vertical motion.

Revision Tips:

- Use vertical motion equations only.
- Max height: vertical velocity = 0.
- Acceleration = $-9.8\,\mathrm{m\,s^{-2}}$ (downwards).